

Takku Lee

Open to Relocate | 2678766194 | alvinbluy@gmail.com

EDUCATION

Northeastern University

Master of Science in Computer Science

Sep 2023 - Aug 2026

Seattle, WA

- **GPA:** 3.9/4, teaching assistant for Object-oriented programming and Algorithm
- **Coursework:** Natural Language Processing, Computer Vision, Computer Networks, Operating Systems, Database, Robotics

Nankai University

Bachelor of Science, Mathematics

Tianjin, China

EXPERIENCE

AI/ML Engineer Intern

TechX

Oct 2025 - Dec 2025

Redmond, WA

- Constructed an AI-powered **iOS** chat app in **Swift** and **Python** tailored for older adults, enhancing user engagement through genAI for natural, empathetic conversations with strong long-term memory for effective personalization.
- Developed end-to-end backend services using **AWS**, including **DynamoDB** for memory persistence, **Lambda** for serverless API logic, and **Kafka** for real-time communication, improving the app's responsiveness and scalability.
- Automated more than 70% of codebase generation using **Vibe Coding** with **Claude Code** and custom **Cursor** rules, improving development speed and ensuring consistency with **MVVM** architecture for enhanced reliability.

Machine Learning Engineer Intern

Berkshire Hathaway HomeServices

Jun 2024 - Aug 2024

Seattle, WA

- Engineered a prediction system with deep learning models using **PyTorch**, improving a prediction error by 30%.
- Designed **RESTful APIs** with **Flask** for model **inference**, integrating seamlessly with the system to support service interactions.
- Developed performance metric dashboards using Jupyter Notebook with **pandas** and Plotly for interactive data visualization.
- Applied **Agile Scrum** practices to optimize data pipelines, reducing development cycle time by 25% through automation.

PhD student in information system and quantitative marketing, Research Assistant

Temple University

Sep 2021 - May 2023

Philadelphia, PA

- Examined the impact of streaming comments on viewers' video appreciation (engagement, perceived quality) in Python and R, applied **causal inference** to design a quasi-experiment, enhancing understanding of viewer interaction patterns.
- Processed 10+GB of **multimodal** data to synchronize streaming comments with video content, and developed a high-performance **NLP** pipeline leveraging **transformer** models for topic classification and sentiment analysis to generate scalable, accurate text insights.

PROJECTS

Game Recommendation with agentic RAG

- Developed an **agentic RAG** pipeline for personalized game recommendations using **LangChain** and **LangGraph**, leveraging user comments to infer individual preferences and recommend games aligned with their interests and playstyle.
- Dynamically orchestrated multi-step retrieval and reasoning, deciding when to query **FAISS** for public reviews, incorporate player preferences, or request **LLM summarization**.
- Deployed via **MCP** and **vLLM** for scalable **GPU inference** with **OpenAI**-compatible API, achieving high-throughput generation and sub-second latency using paged attention and continuous batching.

AI-Driven Flashcard Generator | Qualcomm & Microsoft AI Hackathon Project

- Built flashcard generator to help users efficiently learn and retain new knowledge in edge devices with **FastAPI**, leveraging LLMs for context-awareness within an **MVC** backend. Implemented quantized **Llama-v3.1-8B** using **ONNX**, reducing model size by 70%.
- Proposed energy-efficient **LLM inference** pipeline on edge devices, optimizing for privacy and battery consumption.
- Performed stateless **chunking** and **KV cache** strategies, significantly reducing latency and stabilizing memory usage.

LLM-Driven Robot

- Designed a ROS 2 TurtleBot pipeline on **Linux** for reliable indoor loop traversal within a building, including autonomous return-to-dock behavior, achieving 30 minutes of uninterrupted navigation without collisions.
- Engineered **multi-threaded C++** and Python ROS nodes, integrating Nav2 path planning and real-time motor/sensor firmware, configured DDS networking for multi-node communication over **TCP/UDP**, and interfaced with microcontrollers.
- Implemented **shell scripting** for deployment and debugging, bridging high-level planning with real-time hardware control.

TECHNICAL SKILLS

- **Languages & Databases:** Python(5yrs), Triton, Golang, C/C++, SQL, MySQL, Redis, PostgreSQL, MongoDB, FAISS, Redshift
- **Frameworks & Tools:** PyTorch, Flask, Spring Boot, Express, Angular, FastAPI, Django, Dask, Kubernetes, Terraform, Spark, Linux
- **Cloud:** AWS(Certified Solutions Architect Associate), Azure (Certified AI Engineer Associate), CI/CD, GitHub Actions
- **AI & Machine Learning:** Large Language Models, Prompt Engineering, Natural Language Processing, Computer Vision, AI infra, RAG, Reinforcement Learning